

## Benchmark

**MA.912.GR.3.3** Use coordinate geometry to solve mathematical and real-world geometric problems involving lines, circles, triangles and quadrilaterals.

*Example:* The line  $x + 2y = 10$  is tangent to a circle whose center is located at  $(2, -1)$ . Find the tangent point and a second tangent point of a line with the same slope as the given line.

*Example:* Given  $M(-4, 7)$  and  $N(12, -1)$ , find the coordinates of point  $P$  on  $\overline{MN}$  so that  $P$  partitions  $\overline{MN}$  in the ratio 2:3.

### Benchmark Clarifications:

*Clarification 1:* Problems involving lines include the coordinates of a point on a line segment including the midpoint.

*Clarification 2:* Problems involving circles include determining points on a given circle and finding tangent lines.

*Clarification 3:* Problems involving triangles include median and centroid.

*Clarification 4:* Problems involving quadrilaterals include using parallel and perpendicular slope criteria.

## Connecting Benchmarks/Horizontal Alignment

- MA.912.GR.1.3, MA.912.GR.1.4, MA.912.GR.1.5
- MA.912.GR.7.2

## Terms from the K-12 Glossary

- Circle
- Diameter
- Quadrilateral
- Radius
- Slope
- Triangle

## Vertical Alignment

### Previous Benchmarks

- MA.8.AR.3
- MA.8.GR.1.2
- MA.912.AR.2.2, MA.912.AR.2.3
- MA.912.AR.9.1

### Next Benchmarks

- MA.912.GR.7
- MA.912.T.4

## Purpose and Instructional Strategies

In grade 8 and Algebra 1, students used coordinate systems to study lines and the find distances between points. In Geometry, students expand on their knowledge of coordinate geometry to solve problems geometric problems in real-world and mathematical contexts. In later courses, coordinates will be used to solve a variety of problems involving many shapes, including conic sections and shapes that can be studied using polar coordinates.

- Problem types include finding the midpoint of a segment (midpoint formula); partitioning a segment given endpoints and a ratio; writing the equation of a line, including lines that are parallel or perpendicular; finding the coordinates of the centroid of a triangle; and finding the distance between two points. In some cases, students may need to utilize systems of equations in order to determine solutions.
- Instruction includes the definition of a tangent to a circle and its properties, and the definition of the medians of a triangle and their point of concurrency (centroid).

- Instruction includes various approaches when finding the coordinates of a point partitioning a directed line segment (given the endpoints). (*MTR.2.1, MTR.3.1*) Different methods are described below.
  - The first concept that can be discussed is the connection to weighted average of two points. If the given ratio is  $a:b$ , that means the weights of the endpoints  $(x_1, y_1)$  and  $(x_2, y_2)$  are  $\frac{b}{a+b}$  and  $\frac{a}{a+b}$ , respectively.
    - For example, if the given ratio is 2:3 and the points are at  $A(-3, 6)$  and  $B(4, -8)$  discuss with students where  $P$  is on its way from  $A$  to  $B$ . Students should be able to come with  $\frac{2}{5}$ . That means the weight of  $A$  is  $\frac{3}{5}$  and the weight of  $B$  is  $\frac{2}{5}$ . The next step is to calculate the  $x$ -coordinate of  $P$  using the weighted averages:  $x_P = \frac{3}{5}x_1 + \frac{2}{5}x_2$  which is equivalent to  $x_P = \frac{3}{5}(-3) + \frac{2}{5}(4)$  which is equivalent to  $x_P = -\frac{1}{5}$ . Then calculate the  $y$ -coordinate of  $P$  using the weighted averages:  $y_P = \frac{3}{5}y_1 + \frac{2}{5}y_2$  which is equivalent to  $y_P = \frac{3}{5}(6) + \frac{2}{5}(-8)$  which is equivalent to  $y_P = \frac{2}{5}$ . Therefore,  $P$  is at  $(-\frac{1}{5}, \frac{2}{5})$ .
  - The second method uses the computations of a fraction of the horizontal and the vertical distance between the endpoints (partial distances). That is, if  $P$  is partitioning the segment in the ratio  $a:b$  or  $\frac{a}{a+b}$  of the way from  $(x_1, y_1)$  to  $(x_2, y_2)$ , then its location is  $\frac{a}{a+b}$  of the horizontal distance and  $\frac{a}{a+b}$  of the vertical distance from  $(x_1, y_1)$  to  $(x_2, y_2)$ .
    - For example, if the given ratio is 2:3 and the points are at  $A(-3, 6)$  and  $B(4, -8)$  discuss with students where  $P$  is on its way from  $A$  to  $B$ . Students should be able to determine the horizontal distance from  $A$  to  $B$  as  $x_2 - x_1 = 4 - (-3)$ , which is 7. Then, determine the vertical distance as  $y_2 - y_1 = -8 - 6$ , which is  $-14$ . Since the ratio is 2:3,  $P$  is  $\frac{2}{5}$  of the way from  $A$  to  $B$ . Students can calculate  $\frac{2}{5}$  of the horizontal and the vertical distance,  $\frac{2}{5}(7) = \frac{14}{5}$  and  $\frac{2}{5}(-14) = -\frac{28}{5}$ , respectively. Students should realize that they will need to add the partial distances to  $A$ . Therefore, the coordinates of  $P$  are  $(-3 + \frac{14}{5}, 6 - \frac{28}{5}) = (-\frac{1}{5}, \frac{2}{5})$ .
  - Other methods include the use of formulas. It is important to note that this would require students to memorize formulas and not encourage students to explore all of the concepts.
- Instruction include that understanding that the midpoint of a segment partitions that segment in the ratio 1:1.
- Instruction includes various approaches when finding medians or centroids of triangles. (*MTR.2.1, MTR.3.1*) Different methods are described below.
  - The equation of the line containing a median can be written from the coordinates of the vertex and the coordinates of the midpoint of the opposite side. Solving a system of equations formed by the equations of the two lines containing

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two medians of a triangle will result in the coordinates of the centroid.

- The centroid formula can be given as  $Centroid = \left( \frac{x_1+x_2+x_3}{3}, \frac{y_1+y_2+y_3}{3} \right)$ , where  $(x_1, y_1)$ ,  $(x_2, y_2)$  and  $(x_3, y_3)$  are the coordinates of the vertices of the triangle. Connections should be made to the fact that the coordinates of the centroid are the means of the coordinates of the vertices.
- The centroid is also the center of gravity of the triangle. Show them with a cardboard triangle that the triangle balances perfectly on its centroid (use a pencil or the tip of your finger).
- A method that makes connections to partitions can be found in the Centroid Theorem. This theorem states that the centroid is  $\frac{2}{3}$  of the distance from each vertex to the midpoint of the opposite side, that is, the centroid partitions the median  $\frac{2}{3}$  of the way from the vertex to the midpoint of the opposite side. This theorem can also be used to find the coordinates of centroid.
  - For example, given a triangle with vertices  $A(x_1, y_1)$ ,  $B(x_2, y_2)$  and  $C(x_3, y_3)$ , the midpoints of the sides will be  $M(a_1, b_1)$ ,  $N(a_2, b_2)$  and  $L(a_3, b_3)$  for  $\overline{BC}$ ,  $\overline{AC}$  and  $\overline{AB}$ , respectively. Applying the Centroid Theorem to the median  $\overline{LC}$ , the centroid is at  $\left( \frac{1}{3}x_3 + \frac{2}{3}a_3, \frac{1}{3}y_3 + \frac{2}{3}b_3 \right)$ .  
Using Substitution property of equality,  $\left( \frac{1}{3}x_3 + \frac{2}{3}\left(\frac{x_1+x_2}{2}\right), \frac{1}{3}y_3 + \frac{2}{3}\left(\frac{y_1+y_2}{2}\right) \right) = \left( \frac{1}{3}x_3 + \frac{1}{3}x_1 + \frac{1}{3}x_2, \frac{1}{3}y_3 + \frac{1}{3}y_1 + \frac{1}{3}y_2 \right)$ . This results in  $Centroid = \left( \frac{x_1+x_2+x_3}{3}, \frac{y_1+y_2+y_3}{3} \right)$ .

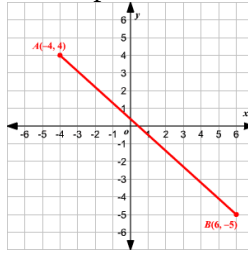
### Common Misconceptions or Errors

- Students may confuse using vertical and horizontal distances when partitioning segments.
- Students may need to be reminded that the segment partition refers to the parts, whose sum is the whole, not a part of a whole.
  - For example, a segment partitioned in a 1:4 ratio is actually 5 parts separated into 1 part and 4 parts, not 1 out of 4 parts.

## Instructional Tasks

### Instructional Task 1 (MTR.3.1)

What are the coordinates of the point that partitions segment  $AB$  in the ratio 2:3?



### Instructional Task 2 (MTR.5.1)

Circle  $A$  has center located at  $(2, 2)$  and contains the point  $(4, 4)$ .

Part A. Write the equation that describes circle  $A$ .

Part B. Write the equation of a line tangent to Circle  $A$  at  $(4, 4)$ .

Part C. Find the equation of a vertical tangent line and of a horizontal tangent line.

### Instructional Task 3 (MTR.3.1)

Triangle  $ABC$  has two of its vertices located at  $(-4, -1)$  and  $(3, -3)$ .

Part A. Triangle  $ABC$  has a centroid located at  $(-1, \frac{1}{3})$ . What is the third vertex of the triangle?

Part B. Determine whether triangle  $ABC$  is a right triangle based on its angle measures and side lengths.

Part C. If triangle  $ABC$  is not a right triangle, can you classify what type of triangle  $ABC$  is?

## Instructional Items

### Instructional Item 1

Given  $J(-4, 2)$  and  $(2, 1)$ , find the coordinates of point  $M$  on  $\overline{JK}$  that partitions the segment into the ratio 1:2.

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*\*The strategies, tasks and items included in the B1G-M are examples and should not be considered comprehensive.*